SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

Gayathri B R
192521330RB

QUESTIONS 25 / 25 completed

Common

COMMON

Write a C program to find the common elements between two arrays.

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
int main()
{
  int a[100],n,i,j,count;
  scanf("%d",&n);
  for(i=0;i<n;i++)
    scanf("%d",&a[i]);
  for(i=0;i<n;i++)
  {
    count=1;
    if(a[i]==-1)
      continue;
    for(j=i+1;j<n;j++)
    {
      if(a[i]==a[j])
      {
```

INPUT

7
2 2 1 5 1 3 3

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C PROGRAMMING

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QUESTIONS 25 / 25 completed

Sum Up Tri.

SUM UP TRI.

Write a C program to find the sum of the upper triangular elements of a matrix.

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
int main()
{
  int a[3][3]={{2,3,4},{5,7,9},{7,8,5}};
  int i,j,sum=0;
  for(i=0;i<3;i++)
    for(j=1;j<3;j++)
      sum+=a[i][j];
  printf("sum of upper triangle =%d",sum);
  return 0;
}
```

INPUT

Your INPUT goes here!

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192

QUESTIONS 25 / 25 completed

Palindrome

PALINDROME

Write a C program to check if an array is a palindrome or not.

PROGRAM EDITOR

C

Run Save

```
#include <stdio.h>

int main()
{
    int a[100], n, i, flag = 1;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for(i = 0; i < n / 2; i++)
    {
        if(a[i] != a[n - 1 - i])
```

INPUT

Enter numbe
elements: 5
Enter elemer
1
2
3
2
1

Search

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C PROGRAMMING

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QUESTIONS 25 / 25 completed

Common

COMMON

Write a C program to find the common elements between two arrays.

PROGRAM EDITOR

C

Run Save

```
#include<stdio.h>
int main()
{
  int a[100],n,i,j,count;
  scanf("%d",&n);
  for(i=0;i<n;i++)
    scanf("%d",&a[i]);
  for(i=0;i<n;i++)
  {
    count=1;
    if(a[i]==-1)
      continue;
    for(j=i+1;j<n;j++)
    {
      if(a[i]==a[j])
      {
```

INPUT

7
2 2 1 5 1 3 3

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QUESTIONS 25 / 25 completed

Add 2 Matrix

ADD 2 MATRIX

Write a C program to add two matrices.

PROGRAM EDITOR

C

Run Save

```
#include <stdio.h>

int main()
{
    int a[10][10], b[10][10], sum[10][10];
    int r, c, i, j;

    printf("Enter rows and columns: ");
    scanf("%d %d", &r, &c);

    printf("Enter first matrix:\n");
    for(i = 0; i < r; i++)
        for(j = 0; j < c; j++)
            scanf("%d", &a[i][j]);

    printf("Enter second matrix:\n");
```

INPUT

Enter rows and columns: 2 2
Enter first matrix:
1 2
3 4
Enter second matrix:
5 6
7 8

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C PROGRAMMING

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QUESTIONS 25 / 25 completed

Multiply Matrix

MULTIPLY MATRIX

Write a C program to multiply two matrices.

PROGRAM EDITOR

C

Run Save

```
#include <stdio.h>

int main()
{
    int a[10][10], b[10][10], c[10][10];
    int r1, c1, r2, c2, i, j, k;

    printf("Enter rows and columns of first matrix: ");
    scanf("%d %d", &r1, &c1);

    printf("Enter rows and columns of second matrix: ");
    scanf("%d %d", &r2, &c2);

    if(c1 != r2)
    {
        printf("Matrix multiplication is not possible.");
```

INPUT

Enter rows and columns of first matrix: 2 2
Enter rows and columns of second matrix: 2 2

Enter first matrix:
1 2
3 4

Enter second matrix:
5 6
7 8

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QUESTIONS 25 / 25 completed

Subtract

SUBTRACT

Write a C program to subtract two matrices.

PROGRAM EDITOR

C

Run Save

```
#include <stdio.h>

int main()
{
    int a[2][2], b[2][2], i, j;

    for(i=0;i<2;i++)
        for(j=0;j<2;j++)
            scanf("%d",&a[i][j]);

    for(i=0;i<2;i++)
        for(j=0;j<2;j++)
            scanf("%d",&b[i][j]);

    for(i=0;i<2;i++)
    {
```

INPUT

```
2 2
5 6
7 8
1 2
3 4
```

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QUESTIONS 25 / 25 completed

Multiply Scalar

MULTIPLY SCALAR

Write a C program to multiply a matrix by a scalar.

PROGRAM EDITOR

C

Run Save

```
#include <stdio.h>

int main()
{
    int a[10][10], r, c, k, i, j;

    scanf("%d%d",&r,&c);
    scanf("%d",&k);

    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&a[i][j]);

    for(i=0;i<r;i++)
    {
        for(j=0;j<c;j++)
```

INPUT

```
2 2
3
1 2
3 4
```

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QUESTIONS 25 / 25 completed

Determinant

DETERMINANT

Write a C program to find the determinant of a 2x2 matrix.

PROGRAM EDITOR

C

Run Save

```
#include <stdio.h>

int main()
{
    int a,b,c,d,det;

    scanf("%d%d%d%d",&a,&b,&c,&d);

    det = a*d - b*c;

    printf("Determinant = %d",det);

    return 0;
}
```

INPUT

```
1 2
3 4
```

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QUESTIONS 25 / 25 completed

Inverse

INVERSE

Write a C program to find the inverse of a 2x2 matrix.

PROGRAM EDITOR C Run Save

```c
#include <stdio.h>

int main()
{
    float a,b,c,d,det;

    scanf("%f%f%f%f",&a,&b,&c,&d);

    det = a*d - b*c;

    if(det == 0)
        printf("Inverse does not exist");
    else
    {
        printf("%.2f %.2f\n",d/det,-b/det);
        printf("%.2f %.2f", -c/det,a/det);
```

INPUT

1 2
3 4

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QUESTIONS 25 / 25 completed

Rank

RANK

Write a C program to find the rank of a matrix.

PROGRAM EDITOR C Run Save

```c
#include <stdio.h>

int main()
{
    int a[10][10], m, n, i, j, k, rank = 0;
    float x;

    scanf("%d%d", &m, &n);

    for(i=0;i<m;i++)
        for(j=0;j<n;j++)
            scanf("%d",&a[i][j]);

    for(i=0;i<m && i<n;i++)
    {
        if(a[i][i] != 0)
```

INPUT

2 2
1 2
2 4

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C PROGRAMMING

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QUESTIONS 25 / 25 completed

Remove

REMOVE

Write a C program to remove an element from an array.

PROGRAM EDITOR C Run Save

```c
#include<stdio.h>
int main()
{
  int a[10],n,i,pos;
  printf("enter the number of elements:");
  scanf("%d",&n);
  printf("enter the array elements:");
  for(i=0;i<)
}
```

INPUT

Your INPUT goes here!

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C PROGRAMMING

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QUESTIONS 25 / 25 completed

Insert in Sorted

INSERT IN SORTED

Write a C program to insert an element in a sorted array.

PROGRAM EDITOR C Run Save

```c
//cyclically rotate an array by one position to rigth
#include<stdio.h>
int main()
{
  int a[100],n,i,temp;
  scanf("%d",&n);
  for(i=0;i<n;i++)
    scanf("%d",&a[i]);
  temp=a[n-1];
  for(i=n-1;i>0;i--)
    a[i]=a[i-1];
  a[0]=temp;
  for(i=0;i<n;i++)
    printf("%d",a[i]);
  return 0;
}
```

INPUT

```
5
1 2 3 4 5
```

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C PROGRAMMING

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QUESTIONS 25 / 25 completed

RM Duplicate

RM DUPLICATE

Write a C program to remove all the occurrences of an element from an array.

PROGRAM EDITOR C Run Save

```c
//roatea matrix 90 degree
#include<stdio.h>
int main()
{
  int a[3][3],i,j;
  for(i=0;i<3;i++)
  {
    for(j=0;j<3;j++)
    {
      scanf("%d",&a[i][j]);
    }
  }
  for(i=0;i<3;i++)
  {
    for(j=2;j>=0;j--)
    {
```

INPUT

```
1 2 3
4 5 6
7 8 9
```

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C PROGRAMMING

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QUESTIONS 25 / 25 completed

Sum Diagonal

SUM DIAGONAL

Write a C program to find the sum of the diagonal elements of a matrix.

A matrix is a two-dimensional array of numbers arranged in rows and columns. The diagonal elements of a matrix are the

PROGRAM EDITOR C Run Save

```c
#include <stdio.h>

int main()
{
    int a[10][10], n, i, j, sum = 0;

    printf("Enter size of matrix: ");
    scanf("%d", &n);

    printf("Enter matrix:\n");

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
            scanf("%d", &a[i][j]);
    }
```

INPUT

```
Enter size of matrix: 3
Enter matrix:
1 2 3
4 5 6
7 8 9
```

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C PROGRAMMING

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QUESTIONS 25 / 25 completed

Sum Up Tri.

SUM UP TRI.

Write a C program to find the sum of the upper triangular elements of a matrix.

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
int main()
{
  int a[3][3]={{2,3,4},{5,7,9},{7,8,5}};
  int i,j,sum=0;
  for(i=0;i<3;i++)
    for(j=1;j<3;j++)
      sum+=a[i][j];
  printf("sum of upper triangle =%d",sum);
  return 0;
}
```

INPUT

Your INPUT goes here!

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QUESTIONS 25 / 25 completed

Transpose

TRANSPOSE

Write a C program to find the transpose of a matrix.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>

int main()
{
    int a[10][10], t[10][10];
    int r, c, i, j;

    printf("Enter rows and columns: ");
    scanf("%d %d", &r, &c);

    printf("Enter matrix:\n");

    for(i = 0; i < r; i++)
        for(j = 0; j < c; j++)
            scanf("%d", &a[i][j]);

```

INPUT

Enter rows and columns: 2 3
Enter matrix:
1 2 3
4 5 6

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